NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 2 EXAMINATION 2024-2025

EE2005 – ELECTRICAL DEVICES AND MACHINES

April / May 2025 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 4 pages.
2. Answer all questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.

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1. Figure 1 shows a magnetic structure with a 500-turn coil wound on the right limb. The mean lengths of the various sections of the core are indicated with $l_1 = 120\ cm$, $l_2 = 9.95\ cm$, $l_3 = 75\ cm$ and an air gap length $l_g = 1\ mm$. The magnetic structure has a uniform cross-sectional area of $25\ cm^2$ throughout.

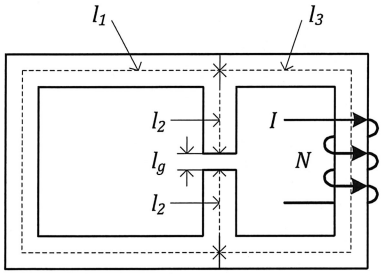


Figure 1

The magnetic characteristics of the core are as follows:

B (T)	0.4	0.75	0.85	1	1.08	1.15	1.175
H (At/m)	125	286	350	500	675	1000	1200

Note: Question No. 1 continues on page 2.

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(a) Draw the magnetic equivalent circuit of the magnetic structure shown in Figure 1 on page 1. Suppose the flux in the air gap (ϕ_g) is 1 mWb. Determine the fluxes in the left limb (ϕ_1) and the right limb (ϕ_3). Also, determine the current in the coil (I) needed to establish these fluxes in the core. Ignore leakage and fringing in your calculations. The permeability of free space is $\mu_0 = 4\pi x10^{-7}$ H/m.

(13 Marks)

(b) Suppose now the relative permeability of the magnetic material $\mu_r = 2526$ is given and the flux in the right limb is $\phi_3 = 1$ mWb. Determine the current in the coil (I) needed to establish this flux.

(8 Marks)

(c) If the current in part (b) is reduced to half its value in 0.5 ms, determine the magnitude of the induced voltage in the coil and the final value of flux ϕ_3.

(4 Marks)

2. Consider the approximate equivalent circuit of a single-phase 80 kVA, 4800/240 V, 50 Hz transformer referred to the primary side where $R_{e1} = 4.5\ \Omega$, $X_{e1} = 8.8\ \Omega$, $R_{c1} = 40.5\ k\Omega$, $X_{m1} = 7.2\ k\Omega$, and $a = 20$.

(a) (i) The secondary of the transformer is short-circuited and a certain voltage is applied to the primary side to give the rated primary current. Determine the primary current, the input power, and the applied voltage. (Hint: Draw the given approximate equivalent circuit to do the calculations.)

(ii) A voltage of 240 V is applied to the secondary of the transformer with the primary side open-circuited. Determine the input power, the current drawn from the 240-V supply, and the input power factor. (Hint: Use the appropriate approximate equivalent circuit to do the calculations.)

(10 Marks)

(b) Using the approximate equivalent circuit of the transformer referred to the secondary side, determine the voltage regulation and the efficiency of the transformer when it delivers full load at 0.8 power factor lagging at rated secondary voltage. Use the rated secondary voltage as the reference, that is, $\vec{V}_2 = 240\angle 0^o\ V$. State any assumption made.

(7 Marks)

(c) The transformer is now connected as an autotransformer supplying 4800 V to a load. Draw the autotransformer connection and label the voltages and currents clearly. Determine the respective percentage of the input power transferred to the load by induction and by conduction. Ignore losses.

(8 Marks)

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3. (a) A three-phase, 50-Hz, 6-pole three-phase induction motor runs at a speed of 960 rpm when delivering an output load torque of 160 Nm. The rotational losses and stator copper losses of the motor are 1200 W and 800 W, respectively.

(i) Determine the rotor copper losses.

(ii) Determine the input power to the motor.

(6 Marks)

(b) A three-phase, 2300-V, 50-Hz, 6-pole, wye-connected induction motor has the following per phase parameters referred to the stator:

$$R_1 = 1.11\ \Omega,\ R_2 = 1.34\ \Omega,\ X_M = 168\ \Omega,\ X_1 = 3.90\ \Omega,\ X_2 = 3.90\ \Omega$$

The rotational losses of the motor are 2672 W. If the motor is running at a slip of 2.95 %, determine

(i) The magnitude of the current drawn by the motor from the supply.
(ii) The torque developed by the motor.
(iii) The output power of the motor.

(12 Marks)

(c) Consider a three-phase, 50-Hz, 6-pole, wye-connected inductor with $R_1 = 0\ \Omega$, $X_1 = 0\ \Omega$ and $R_2 = 0.25\ \Omega$, $X_M = 100\ \Omega$. A maximum torque of 180 Nm is developed by the motor at a speed of 860 rpm. Determine X_2 and the torque developed by the motor at a slip of 4.5 %.

(7 Marks)

4. (a) A dc shunt motor has an armature winding resistance and a field winding resistance of 0.04 Ω and 50 Ω, respectively. The armature terminals are connected to a 250-V dc supply. The rotational losses of the motor are 500 W.

(i) When the motor is running at the rated speed of 1008 rpm, the motor draws a current of 125 A from the supply. Determine the torque developed by the motor, the output power and efficiency of the motor.

(ii) An external variable resistance $R_{ext} = 12.5\ \Omega$ is now connected in series with the field winding and the torque developed is the same as the torque developed in part (i). Determine the new speed of the motor. Assume that the motor is operating in the linear part of the magnetization curve.

(15 Marks)

Note: Question No. 4 continues on page 4.

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(b) A separately-excited dc generator has an armature winding resistance of $0.02\ \Omega$, a field winding resistance of $40\ \Omega$ and rotational losses of 12 kW. The field winding is connected to a 120-V dc supply. The generator is driven by an external prime mover at a constant speed of 800 rpm. When the generator is delivering a current of 800 A to an external load at a terminal voltage of 240 V, determine the input power and output power of the generator. Neglect armature reaction.

(10 Marks)

END OF PAPER

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Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.

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